WHICH FEATURE NETWORK MAKES A GOOD PERCEPTUAL LOSS FOR LOW-DOSE CT DENOISING?

Author(s) Name(s)

Author Affiliation(s)

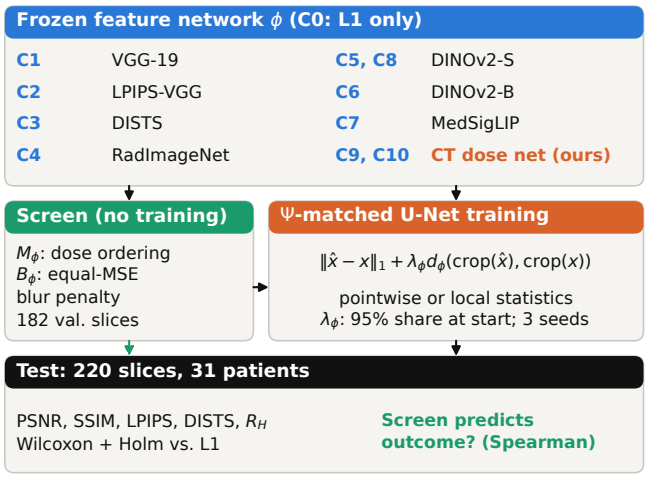


Fig. 1. Only the feature network in the loss changes; its weight is matched by influence, and a screen scores each network before training.

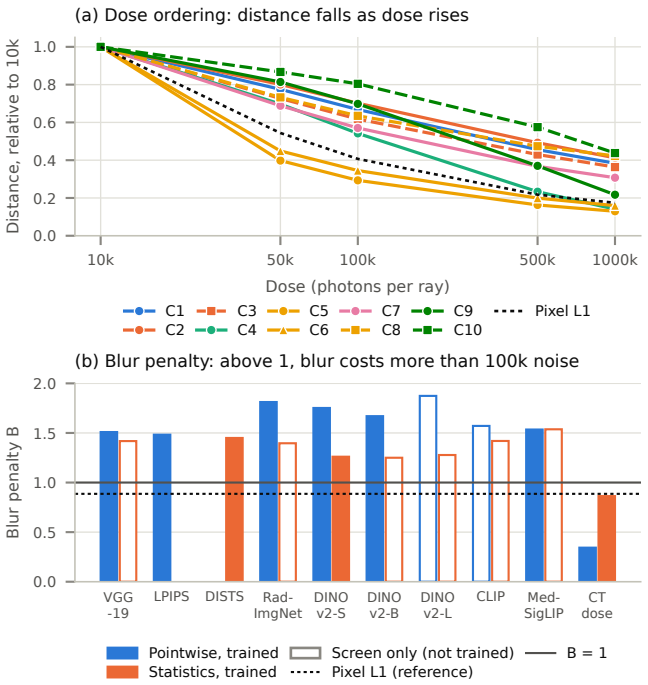


Fig. 2. Training-free screen (validation split). Every pretrained network orders doses almost perfectly (a); only the CT dose network penalizes blur less than noise of the same pixel error (b).

ABSTRACT

TODO (Wed): abstract.

***Index Terms*—** Low-dose CT, denoising, perceptual loss, foundation models, noise texture

1. INTRODUCTION

TODO (Tue): introduction.

2. METHOD

Objective. A U-Net f_θ maps a low-dose slice to $\hat{x}$ and is trained against the normal-dose slice x with

$$\mathcal{L}_\phi = \|\hat{x} - x\|_1 + \lambda_\phi \, d_\phi\big(\mathrm{crop}(\hat{x}), \mathrm{crop}(x)\big),$$

where crop is one random 224×224 crop shared by prediction and target and d_ϕ a distance in the features of a frozen network ϕ. With $\bar{\phi}_l$ the channel-normalized features of block l (C_l channels, $H_l \times W_l$ positions), pointwise matching is

$$d^{\mathrm{pt}}_\phi(\hat{x}, x) = \sum_l \frac{1}{H_l W_l} \sum_{h,w} \big\|\bar{\phi}_l(\hat{x})_{hw} - \bar{\phi}_l(x)_{hw}\big\|_2^2. \qquad (1)$$

CT noise is random, so the output cannot copy the reference grain pixel by pixel. Local statistics matching asks only that each 4×4 window $\omega \in \Omega_l$ of each channel has the same feature mean μ and standard deviation σ:

$$d^{\mathrm{st}}_\phi(\hat{x}, x) = \sum_l \frac{1}{C_l |\Omega_l|} \sum_{c,\omega} \big[1 - s(\hat{\mu}, \mu, c_1)\, s(\hat{\sigma}, \sigma, c_2)\big], \qquad (2)$$

with $s(a, b, c) = (2ab + c)/(a^2 + b^2 + c)$ and $c_1 = c_2 = 10^{-6}$: the texture term of DISTS [1], applied to any network.

Feature networks (Table 2): VGG-19 at relu3_2 [2]; LPIPS-VGG [3] and DISTS with their own weights; RadImageNet ResNet-50, layers 1 to 3 [4]; DINOv2-S and DINOv2-B, blocks 3, 6, 9 [5]; MedSigLIP-448, layers 7, 14, 20 [6]. DINOv2-L and CLIP ViT-B/16 [7] enter the screen only. C8 and C10 use (2), C1, C4 to C7 and C9 use (1).

CT dose network. A loss network should change its features when the noise level changes. Three convolutional blocks (32, 64, 128 channels; 287,328 parameters) with a regression head predict the standardized $\log_{10}$ photon count of 128×128 training-split patches (normal dose fixed at 7), trained with smooth L1 for 10 epochs (lowest validation loss kept) and frozen; its block outputs are the features.

Table 1. Every perceptual loss costs PSNR against L1; the best LPIPS and DISTS values lie within one seed spread of each other, and statistics matching gives DINOv2-S its best texture. Test split (220 slices, 31 patients, 100k). Values: mean±std over seeds 42, 123 and 456. Bold: best trained configuration. † significantly better and ‡ significantly worse than C0 (L1) (paired Wilcoxon on seed-averaged slices, Holm over 30 tests, p < 0.05; PSNR, LPIPS and $|1 - R_H|$ only). * scored with its own training metric. point.: pointwise, stat.: local statistics matching.

ID	Network	Matching	PSNR (dB) ↑	SSIM ↑	LPIPS ↓	DISTS ↓	R_H (→1)	$\lvert 1 - R_H \rvert$ ↓
	LDCT input		23.04	0.7423	0.2843	0.1641	6.174	5.275
C0	–	–	**30.92±0.07**	**0.8691±0.0025**	0.1043±0.0010	0.1112±0.0006	0.657±0.026	0.352±0.026
C1	VGG-19	point.	30.32±0.22‡	0.8606±0.0022	0.0592±0.0010†	0.0668±0.0023	0.798±0.035	0.226±0.030†
C2	LPIPS-VGG	point.	30.39±0.31‡	0.8630±0.0021	0.0563±0.0045$^{\dagger *}$	**0.0645±0.0024**	0.777±0.012	0.242±0.009†
C3	DISTS	stat.	29.81±0.20‡	0.8326±0.0244	**0.0557±0.0066**†	0.0664±0.0014*	0.820±0.014	0.222±0.018†
C4	RadImageNet	point.	28.70±0.32‡	0.8259±0.0075	0.0868±0.0055	0.0847±0.0009	0.798±0.040	0.286±0.017†
C5	DINOv2-S	point.	30.10±0.51‡	0.8625±0.0063	0.0629±0.0078†	0.0707±0.0068	0.682±0.046	0.333±0.044†
C6	DINOv2-B	point.	30.20±0.44‡	0.8618±0.0044	0.0614±0.0027†	0.0713±0.0028	0.731±0.027	0.291±0.029†
C7	MedSigLIP	point.	29.84±0.39‡	0.8596±0.0061	0.0750±0.0114†	0.0737±0.0070	0.531±0.063	0.477±0.060‡
C8	DINOv2-S	stat.	30.26±0.28‡	0.8614±0.0068	0.0579±0.0112†	0.0654±0.0047	0.857±0.017	0.195±0.013†
C9	CT dose net v2	point.	30.44±0.10‡	0.8514±0.0219	0.0673±0.0031†	0.0973±0.0002	0.850±0.018	0.203±0.010†
C10	CT dose net v2	stat.	29.80±0.19‡	0.8475±0.0039	0.0721±0.0063†	0.0982±0.0036	0.951±0.058	**0.167±0.026**†

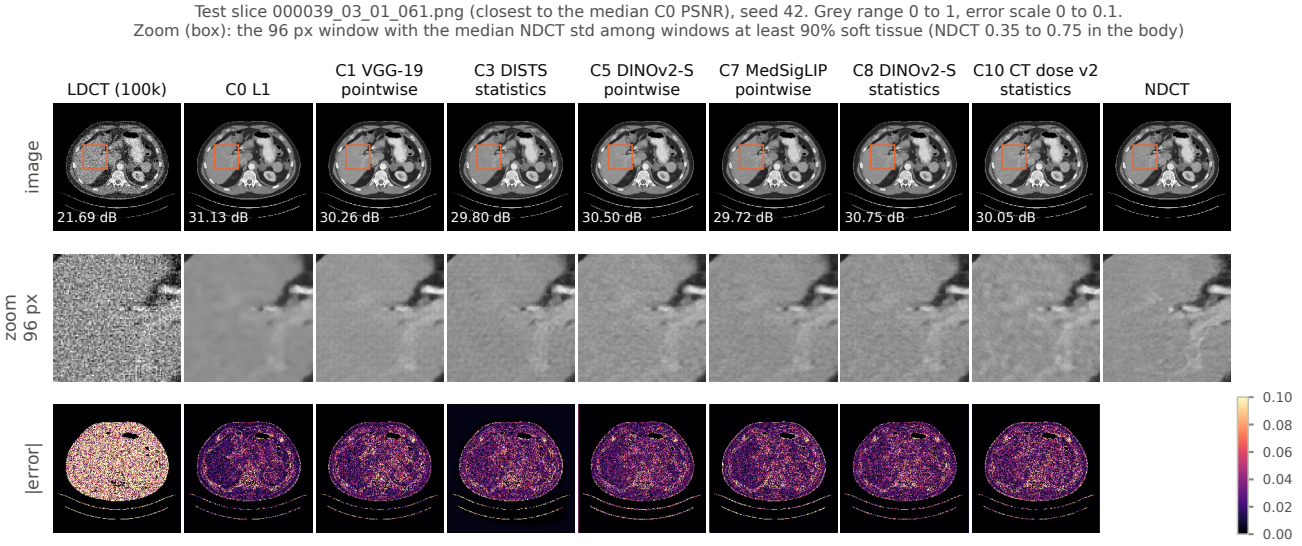


Fig. 3. Test slice closest to the median C0 PSNR, seed 42. Rows: image, zoom, $|\hat{x} - x|$; grey range $[0, 1]$, error scale 0 to 0.1. Zoom: the 96×96 window with the median NDCT standard deviation among windows that are at least 90% soft tissue (NDCT 0.35 to 0.75 inside the body mask). L1 smooths the grain away; statistics matching keeps DINOv2-S grain closer to NDCT than pointwise matching does.

This version replaced a six-class classifier after a validation screen; the adoption threshold on M_ϕ was lowered from 0.9 to 0.8 after seeing the screen, and the runs of the first version are an ablation in the repository. Unlike SACNN [8], its features are trained on dose alone.

Matched influence and screen. As in [9], we set $\lambda_\phi = 19\, S_1/S_\phi$, so the perceptual term is 95% of the loss at initialization (S_1, S_ϕ: summed L1 and feature terms of the untrained U-Net). Before training, each network is screened on the 182 validation slices: the dose ordering M_ϕ is the mean Spearman correlation between $d_\phi(x^{(d)}, x)$ and $\log(1/d)$ over five photon counts d from 10^4 to 10^6, and the blur penalty B_ϕ is the summed distance of a Gaussian blur divided by that of the 10^5 noise at equal MSE. $B_\phi < 1$ means a denoiser can lower the loss by smoothing. Every network is trained regardless of its score.

3. EXPERIMENTS AND RESULTS

Setup. I2I 2025v1 slices from DeepLesion [10] (512×512, $[0, 1]$) with Poisson-simulated low-dose versions [TODO cite]; training input 10^5 photons; patient-level split of 899, 182 and 220 slices (115, 34, 31 patients). All eleven configurations (C0 is L1 only) train the same U-Net [11] with AdamW (10^{-4}), batch 8, 30 epochs and mixed precision, keeping the checkpoint with the lowest validation L1. Seeds 42, 123 and 456 vary the denoiser training only; λ_ϕ and the CT dose network are fixed. Code: TODO(link). Validation L1 in eval mode spikes in some runs (up to 8 of 30 epochs), which leaves checkpoint selection unaffected; one run (C8, seed 42) has no recorded code revision. Metrics: PSNR, SSIM, LPIPS (AlexNet) [3], DISTS, and the high-band noise power ratio R_H at 0.25 to 0.5 cycles per pixel [12], tested as $|1-R_H|$. Family A compares C1 to C10 with C0,

Table 2. Feature networks. Every pretrained network orders doses (M ≥ 0.95); only the CT dose network has B < 1. † feature extractor of the CT dose network v2 (three conv blocks). λ: frozen loss weight. M: dose ordering, B: blur penalty of the training-free screen.

ID	Network	Matching	Params (M)	λ	M ↑	B
C1	VGG-19	point.	1.15	4.699	1.000	1.51
C2	LPIPS-VGG	point.	14.71	9.162	0.998	1.48
C3	DISTS	stat.	14.71	6.984	0.984	1.45
C4	RadImageNet	point.	25.56	2.211	0.954	1.81
C5	DINOv2-S	point.	22.06	1.634	0.998	1.75
C6	DINOv2-B	point.	86.58	2.497	0.997	1.67
C7	MedSigLIP	point.	428.57	3.755	0.999	1.53
C8	DINOv2-S	stat.	22.06	1.966	0.997	1.26
C9	CT dose net v2	point.	0.29†	1.549	0.971	0.34
C10	CT dose net v2	stat.	0.29†	2.802	0.852	0.86

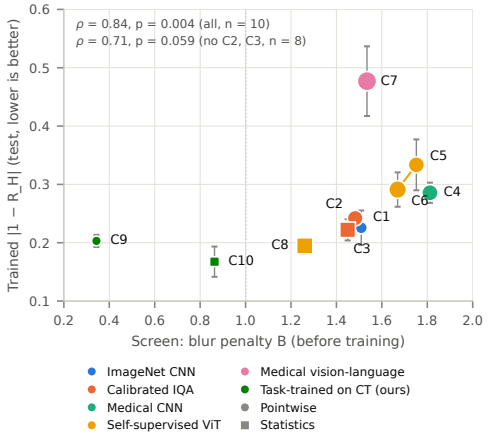


Fig. 4. Blur penalty against trained texture error. The screen does not predict the trained result ($\rho = 0.84$, exploratory).

Family B C8 with C5, C10 with C9 and C6 with C5, on PSNR, LPIPS and $|1 - R_H|$ (Holm over 30 and 9 tests), with paired Wilcoxon tests over the 31 test patients (seeds averaged per slice, slices per patient), effect size r and seed spread; slice-level tests are in the repository. Screen-versus-outcome Spearman correlations are exploratory: C5 and C8, C9 and C10, and C1 to C3 share feature networks.

TODO (Tue): results.

4. CONCLUSION

TODO (Wed): conclusion.

5. COMPLIANCE WITH ETHICAL STANDARDS

This research study was conducted retrospectively using human subject data made available in open access by the NIH Clinical Center (DeepLesion). Ethical approval was not required as confirmed by the license attached with the open access data.

6. ACKNOWLEDGMENTS

TODO: funding and conflicts of interest.

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